Kai Barker

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EDUCATION

University of California, Santa Barbara
Statistics and Data Science B.S.
Major GPA: 3.84/4.0
Relevant Coursework: Machine Learning, Time Series, Stochastic Processes, Advanced Statistical Models, Computational Science, Differential Equations, Technology Management, Probability and Statistics, Problem Solving in Computer Science
Clubs: Consult Your Community, Statistics and Data Science Club, Data Science Collaborative, Ski and Snow Club

WORK EXPERIENCE

Consult Your Community — October 2025- Present
Business Analyst — Santa Barbara, CA

- Conducting business readiness assessment for a small business preparing for acquisition, analyzing operations, financials, and marketing to identify streamlining opportunities and reduce owner dependency
- Analyzing financials, marketing strategies, and operational workflows to position business as turnkey for acquisition
- Developing data-driven recommendations to strengthen operational and financial systems

Department of Statistics and Applied Probability — September 2025- Present
Undergraduate Researcher — Santa Barbara, CA

- Selected for independent research program under faculty sponsorship of Dr. Tomoyuki Ichiba, Department Chair
- Conducting independent research analyzing congressional stock trading patterns and individual performance
- Implementing statistical significance testing methods and data cleaning algorithms to validate trading pattern findings

Foundations of Financial Technology Lab — September 2024- Present
Research Assistant — Santa Barbara, CA

- Analyzed wealth inequality in cryptocurrency markets by examining 100+ million Ethereum wallets using Python
- Implemented a data cleaning pipeline to filter wallets and ensure accurate representation of market participants
- Produced comprehensive visualizations, quantified wealth distribution trends across each blocks top 1 million wallets

PROJECTS

NeurIPS Ariel Data Challenge — July 2024- September 2025
Competition — Santa Barbara, CA

- Developed ensemble machine learning models (XGBoost + Gaussian Process Regression + Ridge) to predict exoplanet atmospheric compositions from 200+ GB of noisy spectral telescope data for ESA's Ariel mission
- Preprocessed and analyzed 14,000 messy image files across 1,100+ exoplanets, implementing advanced signal processing and noise reduction techniques for the observations
- Achieved a GLL score of 0.311 predicting atmospheric signals and uncertainties across 283 wavelength channels

Statistical Modelling for NBA Player Performance — July 2024- October 2024
Personal Project — Santa Barbara, CA

- Developed machine learning models, engineered features to predict player rebounds using time series NBA API data
- Implemented SHAP for model interpretability, significantly improved prediction accuracy over baselines by 20%
- Processed and analyzed performance outcomes to identify trends and validate predictions against real outcomes

SKILLS AND INTERESTS

Programming: Python, R, SQL, JavaScript, Java, C++, CSS, HTML, Typescript

Technical Skills: Microsoft Excel, Tableau, Feature Engineering, Statistical Modelling, Data Visualization, Data Pipeline Development, Git, API Integration, Canva, PowerPoint, Pandas, NumPy, Matplotlib, Scikit-learn

Achievements: HS Valedictorian, Seal of Biliteracy (Spanish), UCSB Dean’s List for Academic Excellence

Interests: Basketball, Hiking, Camping, Backpacking, Cooking, Running, Rock Climbing, Pickleball